

Question 1 (3 marks)

A sequence $\mu_1, \mu_2, \mu_3 \dots \mu_n$ is such that any three consecutive terms are related by the equation $\mu_{n+3} = 6\mu_{n+2} - 5\mu_{n+1}$. It is given that $\mu_1 = 2$ and $\mu_2 = 6$.

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Use mathematical induction to prove that $\mu_n = 5^{n-1} + 1$.

Question 2 (10 marks)

Consider the equation $z^4 - z^3 + z^2 - z + 1 = 0$

(a) Show that the roots of this equations are $e^{\pm \frac{i\pi}{5}}$ and $e^{\pm \frac{i3\pi}{5}}$

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(b) Explain why the roots satisfy $z^2 - z + 1 - \frac{1}{z} + \frac{1}{z^2} = 0$

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(c) By letting $x = z + \frac{1}{z}$, obtain a quadratic equation in x

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(d) Hence find the exact value of $\cos\left(\frac{3\pi}{5}\right)$

3**Question 3 (8 Marks)**

A particle of mass m is thrown from a place h metres above ground level with an initial velocity u at an angle of α to the horizontal.

The particle experiences the effect of gravity, and a resistance proportional to its velocity in both the horizontal and vertical directions are given respectively by

$$\frac{d^2x}{dt^2} = -k \frac{dx}{dt} \text{ and } \frac{d^2y}{dt^2} = -g - k \frac{dy}{dt}$$

Where k is a constant and the acceleration due to gravity is g (You are not required to show these)

(a) Show that $x = \frac{u \cos \alpha}{k} (1 - e^{-kt})$

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(b) Show that the equation of path can be written as

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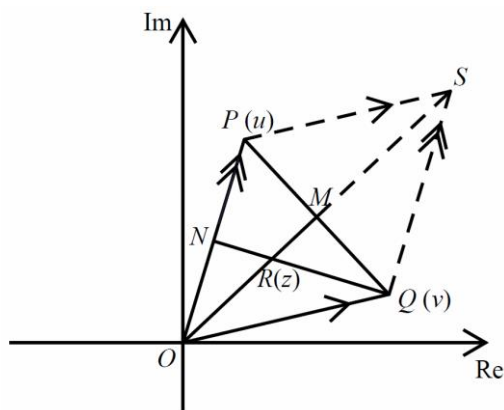
$$y = h + \frac{1}{k} (u \sin \alpha + g \ln \left(-\frac{kx}{u \cos \alpha} + 1 \right) + \frac{1}{k^2} \left(g - (g + ku \sin \alpha) \left(-\frac{xk}{\cos \alpha} + 1 \right) \right)$$

Question 4 (6 marks)

In the Argand diagram below, P and Q represent complex numbers u and v respectively. O , P and Q are not collinear.

In $\triangle OPQ$, the line from O to the midpoint M of PQ , meets the line from Q to the midpoint N of OP in the point R . Let R represent the complex number z .

S is the point such that $OPSQ$ is a parallelogram.



- (a) Explain why there are positive real numbers k and l such that

$$kz = \frac{1}{2}(u + v) \text{ and } l(z - v) = \frac{1}{2}u - v$$

- (b) Hence show $(3 - 2l)v = 2(k - l)z$.

- (c) Deduce that $z = \frac{1}{3}(u + v)$.

Question 5 (6 marks)

Let $T_n(x) = \frac{{}^nC_0}{x} - \frac{{}^nC_1}{x+1} + \frac{{}^nC_2}{x+2} - \dots + (-1)^n \frac{{}^nC_n}{x+n}$ for a given integer n and all real x

- (a) If $S_k(x) = \frac{k!}{x(x+1)(x+2)\dots(x+k)}$ where k is an integer, show that

$$S_k(x) - S_k(x+1) = S_{k+1}(x)$$

- (b) Hence prove using mathematical induction that for $n \geq 1$

$$T_n(x) = \frac{n!}{x(x+1)(x+2)\dots(x+n)}$$

NOTE: you may use without proof the result ${}^{m+1}C_r = {}^mC_r + {}^mC_{r-1}$

End of Examination

Question 1

Test $n=1$

$$\mu_1 = 5^0 + 1$$

$$= 2$$

$\therefore$ true for $n=1$

Test $n=2$

$$\mu_2 = 5^1 + 1$$

$$= 6$$

$\therefore$ true for $n=2$

Assume true for $n=k+1$ and $n=k+2$

i.e. $\mu_{k+1} = 5^k + 1$

$$\mu_{k+2} = 5^{k+1} + 1$$

*

Prove true for $n=k+3$

i.e. $\mu_{k+3} = 5^{k+2} + 1$

Now $\mu_{k+3} = 6\mu_{k+2} - 5\mu_{k+1}$

$$= 6(5^{k+1} + 1) - 5(5^k + 1) \quad [\text{From assumption *}]$$

$$= 6 \cdot 5^{k+1} + 6 - 5^{k+1} - 5$$

$$= 5 \cdot 5^{k+1} + 1$$

$$= 5^{k+2} + 1$$

$\therefore$ true for $n=k+3$

$\therefore$ true for all n , by principle of mathematical induction.

Question 2

a) Consider $z^5 + 1 = (z+1)(z^4 - z^3 + z^2 - z + 1)$

The roots of $z^5 + 1 = 0$ are the fifth roots of -1

$$z = e^{i(\pi + 2k\pi)}, \quad k = 0, \pm 1, \pm 2$$

$$\therefore z = e^{\frac{i}{5}(\pi + 2k\pi)}$$

$$\therefore z = e^{i\frac{\pi}{5}}, e^{i\frac{3\pi}{5}}, e^{i\frac{5\pi}{5}}, e^{i\frac{7\pi}{5}}, e^{i\frac{9\pi}{5}} \text{ are the roots of } z^5 + 1 = 0$$

$$z^5 + 1 = 0 \Rightarrow (z+1) = 0 \text{ or } (z^4 - z^3 + z^2 - z + 1) = 0$$

$$z = e^{i\frac{5\pi}{5}} = -1 \text{ corresponds to } z+1=0$$

$$\therefore \text{the roots of } z^4 - z^3 + z^2 - z + 1 = 0 \text{ are } z = e^{i\frac{\pi}{5}}, e^{i\frac{3\pi}{5}}$$

b) Consider $z^4 - z^3 + z^2 - z + 1 = z^2 \left(z^2 - z + 1 - \frac{1}{z} + \frac{1}{z^2} \right)$

$$z^4 - z^3 + z^2 - z + 1 = 0 \Rightarrow z = 0 \text{ or } z^2 - z + 1 - \frac{1}{z} + \frac{1}{z^2} = 0$$

$$z = 0 \text{ is not a root of } z^4 - z^3 + z^2 - z + 1 = 0$$

$$\text{Thus the roots of } z^4 - z^3 + z^2 - z + 1 = 0 \text{ are also the roots of } z^2 - z + 1 - \frac{1}{z} + \frac{1}{z^2} = 0$$

c) $x = z + \frac{1}{z}$

$$x^2 = \left(z + \frac{1}{z} \right)^2 = z^2 + \frac{1}{z^2} + 2$$

$$z^2 + \frac{1}{z^2} = \left(z + \frac{1}{z} \right)^2 - 2$$

$$= x^2 - 2$$

$$\text{Substituting into equation } z^2 - z + 1 - \frac{1}{z} + \frac{1}{z^2} = 0$$

$$\left(z^2 + \frac{1}{z^2} \right) - \left(z + \frac{1}{z} \right) + 1 = 0$$

$$x^2 - 2 - x + 1 = 0$$

$$x^2 - x - 1 = 0$$

d) The roots of $x^2 - x - 1 = 0$ corresponds to $z^2 - z + 1 - \frac{1}{z} + \frac{1}{z^2} = 0$ where

$$z + \frac{1}{z} = x$$

$$x = e^{i\theta} + \frac{1}{e^{i\theta}} = 2\cos\theta$$

$$\therefore x = 2\cos\frac{\pi}{5}, 2\cos\frac{3\pi}{5} \text{ as } \cos\theta \text{ is even}$$

using quadratic formula

$$x^2 - x - 1 = 0 \text{ has roots } x = \frac{1 \pm \sqrt{5}}{2}$$

as $\frac{3\pi}{5}$ is in quadrant 2

$$\cos\frac{3\pi}{5} < 0$$

$$\therefore 2\cos\frac{3\pi}{5} = \frac{1 - \sqrt{5}}{2}$$

$$\cos\frac{3\pi}{5} = \frac{1 - \sqrt{5}}{4}$$

Question 3

a) using the given equation of motion

$$\frac{d^2x}{dt^2} = -k \frac{dx}{dt} \quad \text{and} \quad \frac{d^2y}{dt^2} = -g - k \frac{dy}{dt}$$

solving these DEs with initial condition

$$x(0) = 0$$

$$y(0) = h$$

$$x'(0) = u \cos \alpha$$

$$y'(0) = u \sin \alpha$$

using separation of variables

$$\frac{dv_x}{dt} = -k v_x$$

$$\int \frac{dv_x}{v_x} = \int -k dt$$

$$\ln v_x = -kt + C_1$$

$$v_x = C_2 e^{-kt}$$

$$\text{when } t=0 \quad v_x = u \cos \alpha \quad \therefore C_2 = u \cos \alpha$$

$$v_x = u \cos \alpha e^{-kt}$$

$$\therefore x = -\frac{u \cos \alpha}{k} e^{-kt} + C_3$$

$$\text{when } t=0 \quad x=0 \quad \therefore C_3 = \frac{u \cos \alpha}{k}$$

$$\therefore x = \frac{u \cos \alpha}{k} (1 - e^{-kt})$$

b) Similarly for y,

$$\frac{dv_y}{dt} = -g - k v_y$$

$$\frac{dv_y}{-g - k v_y} = dt$$

$$\frac{1}{k} \ln(g + k v_y) = -t + D$$

$$g + k v_y = D_1 e^{-kt}$$

$$\text{when } t=0 \quad v_y = u \sin \alpha$$

$$\therefore D_1 = g + k(u \sin \alpha)$$

$$\therefore g + k v_y = (g + k u \sin \alpha) e^{-kt}$$

$$\therefore \frac{dy}{dt} = \frac{1}{k} (-g + (g + k u \sin \alpha) e^{-kt})$$

using the initial conditions $y(0) = h$

$$y = -\frac{g}{k}t - \frac{ge^{-kt}}{k^2} - \frac{u \sin \alpha e^{-kt}}{k} + b_2$$

$$b_2 = h + \frac{g}{k^2} + \frac{u \sin \alpha}{k}$$

$$\begin{aligned}\therefore y &= -\frac{g}{k}t - \frac{ge^{-kt}}{k^2} - \frac{u \sin \alpha e^{-kt}}{k} + h + \frac{g}{k^2} + \frac{u \sin \alpha}{k} \\ &= h + \frac{1}{k} (u \sin \alpha - gt) + \frac{1}{k^2} (g - ge^{-kt} - uk \sin \alpha e^{-kt}) \\ &= h + \frac{1}{k} (u \sin \alpha - gt) + \frac{1}{k^2} (g - e^{-kt} (g + k u \sin \alpha))\end{aligned}$$

$$\text{Now } x = \frac{u \cos \alpha}{k} (1 - e^{-kt})$$

$$\therefore t = -\frac{1}{k} \ln \left(\frac{-xk}{u \cos \alpha} + 1 \right)$$

$$\therefore y = h + \frac{1}{k} \left(u \sin \alpha + g \left(\frac{-xk}{u \cos \alpha} + 1 \right) \right) + \frac{1}{k^2} \left(g - (g + k u \sin \alpha) \left(\frac{-xk}{u \cos \alpha} + 1 \right) \right)$$

Question 4

a) M corresponds to $\frac{1}{2}(u+v)$

$\vec{OR}$ and $\vec{OM}$ are coincident i.e. $\arg(\vec{OR}) = \arg(\vec{OM})$

So there are real numbers k such that $kz = \frac{1}{2}(u+v)$

Now N corresponds to the complex number $\frac{1}{2}u$ and so the

$\vec{ON}$ corresponds to $\frac{1}{2}u-v$ and $\vec{QR}$ corresponds to $z-v$

as $\vec{QR}$ and $\vec{ON}$ are coincident i.e. $\arg(\vec{QR}) = \arg(\vec{ON})$

there is a real number l such that $l(z-v) = \frac{1}{2}u-v$

b) $kz = \frac{1}{2}u + \frac{1}{2}v$ ①

$lz - lv = \frac{1}{2}u - v$ ②

① - ②

$$kz - lz + lv = \frac{3}{2}v$$

$$2kz - 2lz + 2lv = 3v$$

$$2(k-l)z = (3-2l)v$$

c) From b) $(3-2l)v = 2(k-l)z$

As $3-2l$ and $2(k-l)$ are real numbers this implies that the vectors representing the complex numbers v and z are coincident.

This is impossible as the vectors $\vec{OR}$ and $\vec{ON}$ are not coincident.

$$\therefore 3-2l = 2(k-l) = 0$$

$$\therefore k = l = \frac{3}{2}$$

As $kz = \frac{1}{2}(u+v)$ then $\frac{3}{2}z = \frac{1}{2}(u+v)$

$$\therefore z = \frac{1}{3}(u+v)$$

Question 5

a) LHS = $S_k(x) - S_k(x+1)$

$$= \frac{k!}{x(x+1)(x+2)\dots(x+k)} - \frac{k!}{(x+1)(x+2)(x+3)\dots(x+k+1)}$$

$$= \frac{k! (x+k+1) - xk!}{x(x+1)(x+2)\dots(x+k+1)}$$

$$= \frac{k! ((x+k+1) - x)}{x(x+1)(x+2)\dots(x+k+1)}$$

$$= \frac{k! (k+1)}{x(x+1)(x+2)\dots(x+k+1)}$$

$$= \frac{(k+1)!}{x(x+1)(x+2)\dots(x+k+1)}$$

$$= S_{k+1}(x)$$

b) Test $n=1$

$$\text{LHS} = T_1(x) = \frac{1!}{x} - \frac{1!}{x+1}$$

$$= \frac{1}{x} - \frac{1}{x+1}$$

$$= \frac{1}{x(x+1)}$$

$$\text{RHS} = \frac{1!}{x(x+1)} = \frac{1}{x(x+1)}$$

$\therefore$ true for $n=1$

Assume true for $n=k$ i.e. $T_k(x) = \frac{k!}{x(x+1)(x+2)\dots(x+k)}$

Prove true for $n=k+1$ i.e. $T_{k+1}(x) = \frac{(k+1)!}{x(x+1)(x+2)\dots(x+k+1)}$

$$\text{LHS} = T_{k+1}(x)$$

$$= \frac{k!}{x} C_0 - \frac{k!}{x+1} C_1 + \frac{k!}{x+2} C_2 - \dots + (-1)^k \frac{k!}{x+k} C_k + (-1)^{k+1} \frac{k!}{x+k+1} C_{k+1}$$

$$= \frac{k!}{x} C_0 - \frac{(k! C_1 + k! C_0)}{x+1} + \frac{(k! C_2 + k! C_1)}{x+2} - \dots + (-1)^k \frac{(k! C_k + k! C_{k-1})}{x+k} + (-1)^{k+1} \frac{k! C_k}{(x+1)+k}$$

$$= \frac{k!}{x} C_0 - \frac{k! C_1}{x+1} + \frac{k! C_2}{x+2} - \dots + (-1)^k \frac{k! C_k}{x+k} - \left[\frac{k! C_0}{x+1} - \frac{k! C_1}{x+2} + \dots + (-1)^{k+1} \frac{k! C_{k-1}}{x+k} + (-1)^k \frac{k! C_k}{x+1+k} \right]$$

$$= T_k(x) - T_k(x+1)$$

$$= S_k(x) - S_k(x+1) \quad [\text{from assumption *}]$$

$$= S_{k+1}(x) \quad [\text{from (a)}]$$

$\therefore$ true for $n = k+1$

$\therefore$ true for all $n \geq 1$ by principle of mathematical induction.